



# Modelling Financial Markets

## Chapter 1: Stylised Facts and Returns



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

## Acronyms used in this material (1/2)

- ACF: AutoCorrelation Function
- AI: Artificial Intelligence
- AR: AutoRegressive (model); in event studies: Abnormal Return
- ARCH: AutoRegressive Conditional Heteroskedasticity
- ARCH-LM: ARCH Lagrange Multiplier (test)
- BET-TR: BET Total Return (index)
- BNR: Banca Națională a României (the National Bank of Romania)
- BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- CAGR: Compound Annual Growth Rate
- CI: Confidence Interval
- CLT: Central Limit Theorem
- CRIX: CRypto IndeX
- DAX: Deutscher Aktienindex (the German stock index)
- EGARCH: Exponential GARCH
- ETF: Exchange-Traded Fund
- FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- GEO: Government Emergency Ordinance
- GJR-GARCH: Glosten-Jagannathan-Runkle GARCH
- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- HAR-RV: Heterogeneous AutoRegressive model of Realised Volatility
- JB: Jarque-Bera (test)
- LB: Ljung-Box (test)
- MDD: Maximum DrawDown
- OHLC: Open, High, Low, Close
- PDV: Path-Dependent Volatility

## Acronyms used in this material (2/2)

- ▣ **QQ: Quantile–Quantile (plot)**
- ▣ **RMSE: Root Mean Squared Error**
- ▣ **SE: Standard Error**
- ▣ **TSPL: Time-Shifted Power Law (kernel)**
- ▣ **US: United States**
- ▣ **VIX: CBOE Volatility Index**

## Chapter Route and What a Stylised Fact Is

### Route of this chapter

- Part I: returns, distribution and tails, dependence and its mechanisms
  - Part II: persistence and the limits of inference, market exceptions, OHLC volatility, performance, a case study, one pass of the AI-assisted research loop
  - Technical detail sits in the [Appendix](#) at the end
- 
- Stylised facts are the benchmark every model of returns must reproduce: the i.i.d. (independent and identically distributed) Normal-distribution model fails most of them

### Definition

- A **stylised fact** is a qualitative statistical regularity of **returns** seen across assets, markets and periods: “returns have heavy tails”, not “kurtosis = 7.3” [Cont \(2001\)](#)
- It is a hypothesis to re-test on each market and frequency [Ratliff-Crain, Van Oort, Koehler & Tivnan \(2025\)](#)
- Key message: returns show little linear autocorrelation, but their **size** is highly predictable

## Data Used in This Chapter

| Series  | Price used              | Type, currency                | Calendar                 | Sample          | $T$   |
|---------|-------------------------|-------------------------------|--------------------------|-----------------|-------|
| S&P 500 | daily close (plus OHLC) | price index, USD              | US trading days          | 1990–2026       | 9 245 |
| BET-TR  | daily close             | total return, RON             | BVB trading days         | Sep 2014 – 2026 | 2 992 |
| Bitcoin | daily close             | price, USD                    | every day, 24/7          | Sep 2014 – 2026 | 4 384 |
| EUR/RON | BNR reference rate      | RON per EUR,<br>managed float | Romanian working<br>days | Jul 2005 – 2026 | 5 352 |
| Gold    | daily close             | spot, USD per ounce           | weekdays only            | 1990–2026       | 9 559 |

- Daily closes up to 18 September 2026; daily log returns on each series' own calendar
- **Managed float:** a market exchange rate with central-bank interventions; the official BNR fixing replaces the EUR/RON market (interbank) quotes, whose lag-1 autocorrelation of  $-0.4$  to  $-0.7$  in several years came from quote errors
- S&P 500 opens before 2007 equal the previous close: range-based estimators start in 2008

## Returns: Definitions and Aggregation

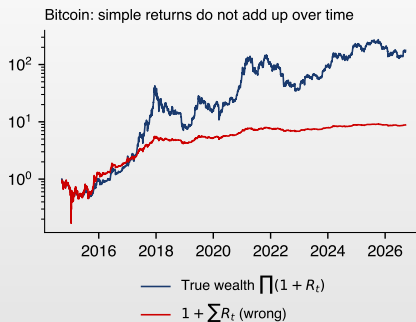
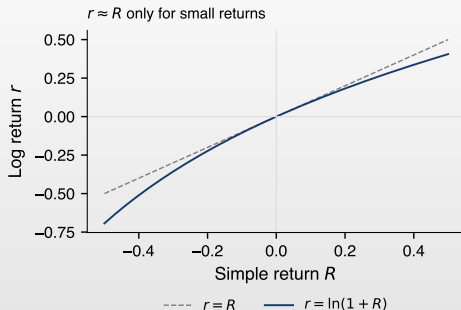
|                  | Simple return $R_t$                    | Log return $r_t$                                     |
|------------------|----------------------------------------|------------------------------------------------------|
| Definition       | $(P_t + D_t)/P_{t-1} - 1$              | $\ln(1 + R_t) = \ln P_t - \ln P_{t-1}$ (no dividend) |
| Over $k$ periods | $\prod_j (1 + R_{t-j}) - 1$            | $\sum_j r_{t-j}$ : additive over time                |
| Across assets    | $R_{p,t} = \sum_j w_j R_{i,t}$ : exact | only approximately                                   |
| Range            | $R_t \geq -100\%$                      | $(-\infty, +\infty)$                                 |
| Use              | portfolios, performance reporting      | single-asset dynamics, CLT arguments                 |

- $P_t$  is the close,  $D_t$  the dividend paid in  $(t-1, t]$ ; with dividends use total-return or adjusted series (BET-TR)
- $\ln(1 + R) < R$  for every  $R \neq 0$ :  $+1\% \rightarrow 0.995\%$ , but  $-50\% \rightarrow -69.3\%$
- Textbooks: [Campbell, Lo & MacKinlay \(1997\)](#), ch. 1; [Franke, Härdle & Hafner \(2019\)](#)

## Annualisation, CAGR and Volatility Drag

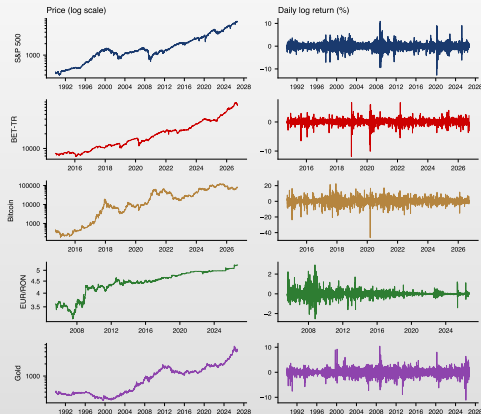
- $P$  = return intervals per year: 252 (stocks), 365 (crypto, traded every day), the actual  $\approx 260$  weekday quotes (gold)
- $\mu_{\text{ann}} = P \bar{r}$  needs only a constant mean;  $\sigma_{\text{ann}} = \sqrt{P} \hat{\sigma}$  is exact only for serially uncorrelated returns with constant variance
  - ▶ with autocorrelation the factor changes: EUR/RON, lag-1 autocorrelation 0.17: 4.4%  $\rightarrow$  5.3% a year (▶ [Appendix](#); Seminar A2)
- **CAGR** (Compound Annual Growth Rate)  $= (P_T/P_0)^{1/Y} - 1$ ,  $Y$  = number of years; the geometric mean  $\approx \mu_{\text{arith}} - \sigma^2/2$  (**volatility drag**: the gap between the arithmetic and the geometric mean)
  - ▶ S&P 500 1990–2026:  $\hat{\sigma} = 1.133\%$  a day, 18.0% a year; CAGR 8.69% (price index, no dividends)
  - ▶ Bitcoin 2014–2026: arithmetic mean 65.2% a year ( $P = 365$ ), CAGR 53.9%: a drag of 11 pp (percentage points)

## Simple vs Log Returns in Practice



- Left:  $r \approx R$  only for small moves
- Right: Bitcoin 2014–2026 multiplied wealth by 176.9, i.e.  $\sum_t r_t = \ln 176.9 = 5.18$ ; adding daily simple returns gives  $1 + \sum_t R_t = 8.83$ : wrong by a factor of 20

## Five Markets: Prices and Returns



- **Left:** prices on log scale, trending and non-stationary
- **Right:** daily log returns fluctuate around a small mean; calm and turbulent periods alternate; extremes cluster (2008, March 2020)
- EUR/RON moves more than 1% on only 1.4% of days
- Bitcoin, 12 March 2020: log return  $-46.5\%$ , a price fall of  $1 - e^{-0.465} = 37.2\%$

## Sample Moments and Their Sampling Error

- ▣  $z_t = (r_t - \bar{r})/\hat{\sigma}$ ; skewness  $\hat{S} = \frac{1}{T} \sum_t z_t^3$ , kurtosis  $\hat{K} = \frac{1}{T} \sum_t z_t^4$ ; Normal distribution:  $S = 0$ ,  $K = 3$ , excess kurtosis  $K - 3 = 0$
- ▣ Mean:  $\text{SE}(\bar{r}) = \sigma/\sqrt{T}$ ; daily means are 10–60 times smaller than  $\hat{\sigma}$ , so the mean is the least precise moment
  - S&P 500:  $\bar{r} = 0.033\% \pm 0.023\%$  a day (95%),  $T = 9,245$
- ▣ i.i.d. data from the Normal distribution:  $\sqrt{T} \hat{S} \rightarrow N(0, 6)$  and  $\sqrt{T}(\hat{K} - 3) \rightarrow N(0, 24)$
- ▣ Any i.i.d. distribution: the delta method gives a Normal limit for  $\hat{S}$  only if  $\mathbb{E}r^6 < \infty$ , for  $\hat{K}$  only if  $\mathbb{E}r^8 < \infty$ ; whether these moments exist is decided by the tail index (Hill estimator, below)

## Three Classical Tests and What They Assume

| Test                     | Statistic                                                         | $H_0$                         | $\chi^2$ limit needs             | Problem here                              |
|--------------------------|-------------------------------------------------------------------|-------------------------------|----------------------------------|-------------------------------------------|
| JB, Jarque & Bera (1980) | $T \left[ \frac{\hat{S}^2}{6} + \frac{(\hat{K}-3)^2}{24} \right]$ | i.i.d., Normal distribution   | $H_0$ itself (all moments exist) | a rejection does not say which part fails |
| LB, Ljung & Box (1978)   | $T(T+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{T-k}$                  | $\rho_1 = \dots = \rho_m = 0$ | i.i.d., not only uncorrelated    | clustering: over-rejects                  |
| ARCH-LM, Engle (1982)    | $TR^2$ , $\hat{\varepsilon}_t^2$ on $q$ lags                      | i.i.d., no ARCH effects       | $\mathbb{E}r^4 < \infty$         | $\alpha \approx 3 < 4$ : size unreliable  |

- ▣ **ARCH** = AutoRegressive Conditional Heteroskedasticity: today's variance depends on past returns;  $\hat{\varepsilon}_t = r_t - \bar{r}$ ,  $q = 5$ ;  $\alpha$  is the tail index (below)
- ▣ We use  $m = 10$ ; 5% critical values:  $\chi_2^2 = 5.99$ ,  $\chi_5^2 = 11.1$ ,  $\chi_{10}^2 = 18.3$
- ▣ The question is not how to run the tests but whether their reference distribution holds for returns

## Worked Example: JB for the S&P 500

- ▣ S&P 500, 1990–2026:  $T = 9,245$ ,  $\hat{S} = -0.364$ ,  $\hat{K} - 3 = 10.88$ 
  - ▶ skewness term  $\hat{S}^2/6 = 0.132/6 = 0.022$ ; kurtosis term  $10.88^2/24 = 118.4/24 = 4.93$
  - ▶  $JB = 9,245 \times (0.022 + 4.93) \approx 45,800$ , against the critical value 5.99
- ▣ Almost all of the statistic comes from the kurtosis term: the problem is the tails, not the asymmetry
- ▣ Under the null all moments exist, so the p-value is valid for the **joint** null “i.i.d. and Normal distribution”; heavy tails make JB grow with  $T$ : this is power, not a broken p-value
- ▣ A rejection does not say which part fails, the marginal distribution or independence; tests of the marginal distribution with HAC variances: [▶ Appendix](#)

## Robust Alternative: Quantile-Based Skewness and Kurtosis

- Hinkley skewness  $\frac{q_{0.95} + q_{0.05} - 2q_{0.5}}{q_{0.95} - q_{0.05}}$ ; Crow–Siddiqui kurtosis  $\frac{q_{0.975} - q_{0.025}}{q_{0.75} - q_{0.25}} - 2.91$ ; both 0 for the Normal distribution [Kim & White \(2004\)](#)
- No moments needed; 95% CI from a moving-block bootstrap (blocks of 20 days, 1,000 resamples), valid for stationary, weakly dependent returns with a positive density at these quantiles

|         | Moment skew. | Hinkley [95% CI]        | Excess kurt. | Crow–Siddiqui [95% CI] |
|---------|--------------|-------------------------|--------------|------------------------|
| S&P 500 | −0.36        | −0.064 [−0.092, −0.041] | 10.9         | 1.54 [1.29, 1.79]      |
| BET-TR  | −1.40        | −0.036 [−0.085, 0.025]  | 17.8         | 1.06 [0.73, 1.43]      |
| Bitcoin | −0.70        | −0.018 [−0.069, 0.032]  | 12.0         | 2.25 [1.88, 2.64]      |
| EUR/RON | 0.73         | 0.041 [−0.022, 0.093]   | 15.1         | 4.87 [3.74, 6.13]      |
| Gold    | −0.23        | 0.004 [−0.025, 0.030]   | 9.3          | 1.47 [1.29, 1.73]      |

- Excess quantile kurtosis survives in all five markets: no CI contains 0
- Quantile-based negative skewness is established only for the S&P 500; BET-TR's moment skewness −1.40 comes from a few days (19 December 2018)

## Summary Statistics: Five Markets

|                           | S&P 500      | BET-TR      | Bitcoin      | EUR/RON    | Gold         |
|---------------------------|--------------|-------------|--------------|------------|--------------|
| Observations              | 9245         | 2992        | 4384         | 5352       | 9559         |
| Mean (% per day)          | 0.033        | 0.077       | 0.118        | 0.007      | 0.025        |
| Std. dev. (% per day)     | 1.13         | 0.98        | 3.49         | 0.28       | 1.01         |
| Annualised volatility (%) | 18.0         | 15.5        | 66.6         | 4.4        | 16.3         |
| Skewness                  | -0.36        | -1.40       | -0.70        | 0.73       | -0.23        |
| Excess kurtosis           | 10.9         | 17.8        | 12.0         | 15.1       | 9.3          |
| Min / max log return (%)  | -12.8 / 11.0 | -11.9 / 6.7 | -46.5 / 22.5 | -2.5 / 2.9 | -11.1 / 10.4 |
| Jarque-Bera               | 45805        | 40433       | 26482        | 51575      | 34732        |
| Ljung-Box $Q(10), r_t$    | 91.6         | 70.0        | 20.2         | 221.7      | 32.1         |
| Ljung-Box $Q(10), r_t^2$  | 8018         | 714         | 213          | 2420       | 1049         |
| ARCH-LM (5 lags)          | 2229         | 371         | 115          | 871        | 483          |

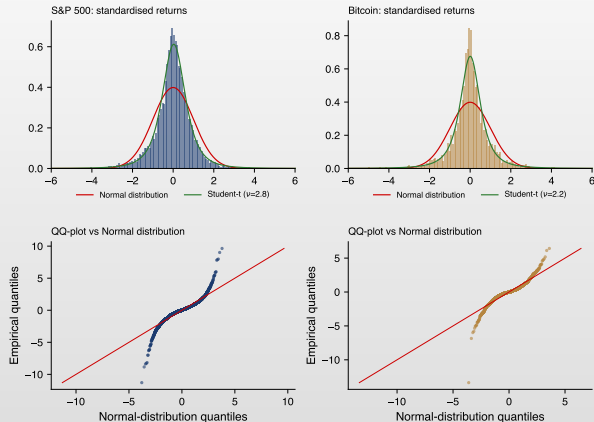
- Descriptive: heavy tails everywhere (excess kurtosis 9.3–17.8);  $Q(10)$  on  $r_t^2$  is 10–90 times larger than on  $r_t$ : the size of returns is far more predictable than their direction
- Inferential: all i.i.d. p-values are below 0.03, but they are valid only under the assumptions of the previous table

## The Stylised Facts of Cont (2001)

| Group                    | Facts                                                                                                                      |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Distribution             | 2. heavy tails; 3. gain/loss asymmetry; 4. aggregational Gaussianity; 7. conditional heavy tails                           |
| Dependence of returns    | 1. absence of linear autocorrelation                                                                                       |
| Dependence of volatility | 5. intermittency; 6. volatility clustering; 8. slow decay of the autocorrelation of $ r_t $ ; 11. asymmetry in time scales |
| Links to other variables | 9. leverage effect; 10. volume/volatility correlation                                                                      |

- We re-test them on five markets, with data up to September 2026, adding the inference each test needs
- Early evidence: Mandelbrot (1963), Fama (1965); textbook treatment: Taylor (2008), Franke, Härdle & Hafner (2019)

## Fact 2, Heavy Tails: Histograms and QQ-Plots



- Extreme returns are far more frequent than under a Normal distribution with the same variance
- Standardised returns  $(r_t - \bar{r})/\hat{\sigma}$ : higher peak and fatter tails
- QQ-plots (empirical quantiles against those of the Normal distribution) bend away from the line in both tails
- A Student- $t$  fits much better

## How Heavy Are the Tails? Exceedances and the Student- $t$

|                                                                  | S&P 500 | BET-TR | Bitcoin | EUR/RON | Gold  |
|------------------------------------------------------------------|---------|--------|---------|---------|-------|
| Days beyond $3\hat{\sigma}$ (%)                                  | 1.55    | 1.44   | 1.82    | 2.06    | 1.42  |
| Days beyond $5\hat{\sigma}$ (count)                              | 30      | 12     | 10      | 29      | 26    |
| Expected under the Normal distribution, $> 5\hat{\sigma}$        | 0.005   | 0.002  | 0.003   | 0.003   | 0.005 |
| Student- $t$ degrees of freedom $\hat{\nu}$ (maximum likelihood) | 2.8     | 3.1    | 2.2     | 1.2     | 2.9   |

- Normal distribution: 0.27% of days beyond  $3\sigma$  (observed 5–8 times more); one  $5\sigma$  day every 6,900 years of trading (observed 10–30)
- Student- $t$ , [Praetz \(1972\)](#):  $f(x) \propto (1 + \frac{(x-\mu)^2}{\nu s^2})^{-(\nu+1)/2}$ ; tail  $P(X > x) \propto x^{-\nu}$ , moments exist only of order  $< \nu$ ; variance  $s^2\nu/(\nu - 2)$  for  $\nu > 2$ , excess kurtosis  $6/(\nu - 4)$  for  $\nu > 4$
- $\hat{\nu}$  is fitted by **maximum likelihood** to the whole distribution: the many small returns dominate, so a misfit centre moves  $\hat{\nu}$ ; EUR/RON's 1.2 versus a tail-only Hill index of 2.8 signals misspecification (57% of daily changes below 0.1%, plus rare jumps)

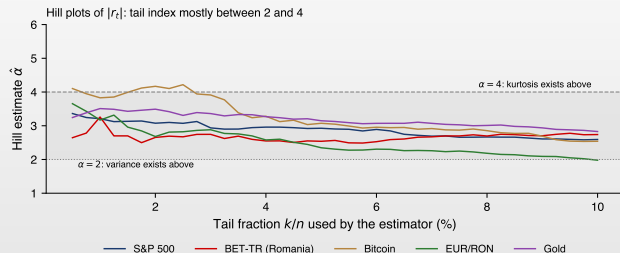
## The Hill Estimator of the Tail Index

- **Tail index**  $\alpha$ :  $P(|X| > x) \approx c x^{-\alpha}$  for large  $x$ ;  $\mathbb{E}|X|^p < \infty$  only for  $p < \alpha$ 
  - ▶  $\alpha > 2$ : finite variance;  $\alpha > 4$ : finite kurtosis; power laws were first argued by Mandelbrot (1963) and Fama (1965)
- **Hill estimator**, Hill (1975): order  $X_{(1)} \geq X_{(2)} \geq \dots \geq X_{(n)}$  and use the  $k$  largest

$$\hat{\alpha}_k = \left[ \frac{1}{k} \sum_{i=1}^k \ln X_{(i)} - \ln X_{(k+1)} \right]^{-1}, \quad \text{SE}(\hat{\alpha}_k) \approx \hat{\alpha}_k / \sqrt{k}$$

- The SE assumes independent tail observations and a Pareto tail; clustering of extremes makes it larger (▶ Appendix)
- **Choice of  $k$** : small  $k$  is noisy, large  $k$  lets the centre in and biases  $\hat{\alpha}$ ; read the level where the **Hill plot** ( $\hat{\alpha}_k$  against  $k$ ) is flat, here  $k = 1\text{--}5\%$  of  $n$

## Hill Plots for Five Markets



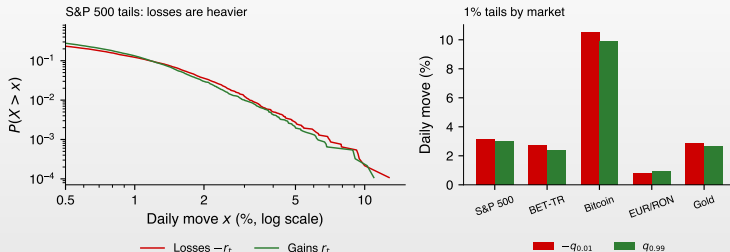
- $\hat{\alpha}$  for  $|r_t|$  at  $k = 2.5\%$  of  $n$ : S&P 500 3.1, BET-TR 2.7, Bitcoin 4.2, EUR/RON 2.8, gold 3.3
- For  $k = 1-5\%$ : between 2.3 and 3.5 except Bitcoin (3.1–4.2)
- Loss tails heavier for BET-TR (2.2 vs 3.1) and Bitcoin (2.7 vs 3.4)
- Tails are modelled with extreme value theory ([Chapter 7](#))

## Tail Indices with Confidence Intervals

| $ r_t $ , $k = 2.5\%$ of $n$ | $n$  | $k$ | $\hat{\alpha}$ | SE i.i.d. | 95% CI (i.i.d.) | SE block bootstrap |
|------------------------------|------|-----|----------------|-----------|-----------------|--------------------|
| S&P 500                      | 9245 | 231 | 3.07           | 0.20      | [2.68, 3.47]    | 0.33               |
| BET-TR                       | 2992 | 74  | 2.67           | 0.31      | [2.06, 3.28]    | 0.45               |
| Bitcoin                      | 4384 | 109 | 4.22           | 0.40      | [3.43, 5.01]    | 0.47               |
| EUR/RON                      | 5352 | 133 | 2.82           | 0.24      | [2.34, 3.30]    | 0.28               |
| Gold                         | 9559 | 238 | 3.31           | 0.21      | [2.89, 3.73]    | 0.24               |

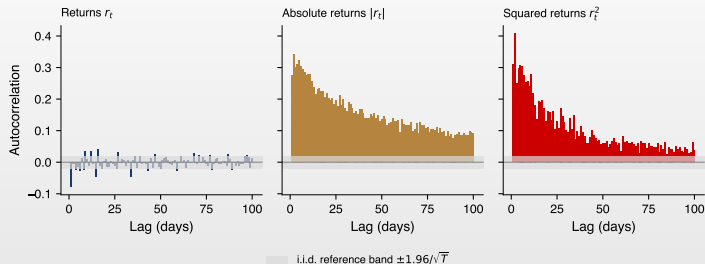
- Moving-block bootstrap (blocks of 20 days, 500 resamples) keeps the clustering of extremes: SEs 1.1–1.6 times the i.i.d. ones
- $\alpha = 4$  is outside every interval except Bitcoin's, also with the bootstrap SE: the fourth moment very likely does not exist
- $\alpha = 2$  is outside every i.i.d. interval; with the bootstrap SE only BET-TR reaches it ( $2.67 - 1.96 \times 0.45 = 1.79$ )
- Bitcoin's 4.2 uses a threshold where its Hill plot is not yet flat; at a common threshold its two tails give 2.71 and 3.36 ([Appendix](#))

## Fact 3: Gain/Loss Asymmetry in the Data



- Survival functions  $P(X > x)$  of losses  $(-r_t)$  and gains  $(r_t)$ , log-log scale: large losses are more frequent than gains of the same size
- **VaR** (Value-at-Risk) 1% =  $-q_{0.01}$ , the loss exceeded on 1% of days (Chapter 7); it exceeds the 99% return quantile  $q_{0.99}$  for 4 of 5 markets, e.g. S&P 500 3.15% vs 2.97%
- EUR/RON is the exception: leu depreciations (up-moves) are larger (0.94% vs 0.76%)
- A Normal-distribution model with the right variance underestimates such quantiles: stress tests based on “ $3\sigma$  events” are too optimistic

## Facts 1 and 6: Uncorrelated but Not Independent



- Fact 1:  $\text{corr}(r_t, r_{t-k}) \approx 0$ ; S&P 500 1990–2026, lag 1:  $-0.08$ ; grey band: i.i.d.  $\pm 1.96/\sqrt{T}$
- Fact 6, volatility clustering:  $\text{corr}(|r_t|, |r_{t-k}|) > 0$  for many lags (lag 1:  $0.27$ ): large moves follow large moves of either sign
- Independence would require zero correlation of **every** function of returns: returns are uncorrelated but not independent; model  $r_t = \sigma_t \varepsilon_t$  with  $\sigma_t$  driven by past returns (GARCH, [Chapter 5](#))

## The ACF under Conditional Heteroskedasticity

- **ACF** (AutoCorrelation Function):  $\hat{\rho}_k = \sum_t x_t x_{t-k} / \sum_t x_t^2$ ,  $x_t = r_t - \bar{r}$ , plotted against  $k$
- Uncorrelated returns whose conditional variance moves:  $\hat{\rho}_k \approx N(0, \tau_k)$ ,  $\hat{\tau}_k = \sum_t x_t^2 x_{t-k}^2 / (\sum_t x_t^2)^2$  [Romano & Thombs \(1996\)](#); i.i.d.:  $T\tau_k = 1$ ; clustering:  $T\tau_k > 1$

| Lag 1                           | S&P 500     | BET-TR    | Bitcoin     | EUR/RON    | Gold        |
|---------------------------------|-------------|-----------|-------------|------------|-------------|
| $\hat{\rho}_1$                  | -0.079      | 0.061     | -0.022      | 0.170      | -0.011      |
| $T\hat{\tau}_1$                 | 5.00        | 4.35      | 2.89        | 6.85       | 2.43        |
| $t$ -statistic, i.i.d. / robust | -7.6 / -3.4 | 3.3 / 1.6 | -1.5 / -0.9 | 12.4 / 4.7 | -1.1 / -0.7 |

- Robust portmanteau  $\tilde{Q}(m) = \sum_{k=1}^m \hat{\rho}_k^2 / \hat{\tau}_k$  [Lobato, Nankervis & Savin \(2001\)](#): S&P 500  $Q(10) = 91.6$  but  $\tilde{Q}(10) = 19.0$  ( $p = 0.041$ ); its assumptions: [Appendix](#)
- With robust SEs only the S&P 500 ( $t = -3.4$ ) and EUR/RON ( $t = 4.7$ ) keep a significant lag-1 autocorrelation, economically small

## Why Heavy Tails? Mixtures of Normal Distributions

- ▣ Suppose  $r_t = \sigma_t \varepsilon_t$  with  $\varepsilon_t \sim N(0, 1)$  independent of a random volatility  $\sigma_t$

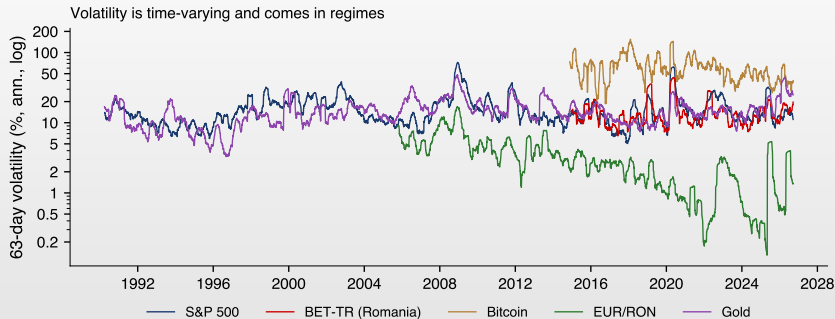
$$K = \frac{\mathbb{E}[r_t^4]}{\mathbb{E}[r_t^2]^2} = 3 \frac{\mathbb{E}[\sigma_t^4]}{\mathbb{E}[\sigma_t^2]^2} \geq 3$$

- ▣ Jensen's inequality ( $\mathbb{E}[X^2] \geq \mathbb{E}[X]^2$ ) with  $X = \sigma^2$ : any variation in volatility creates excess kurtosis
- ▣ Example:  $\sigma = 1\%$  on 90% of days and 3% on 10% of days

$$K = 3 \cdot \frac{0.9 \cdot 1 + 0.1 \cdot 81}{(0.9 + 0.1 \cdot 9)^2} = 3 \cdot \frac{9.0}{3.24} = 8.33$$

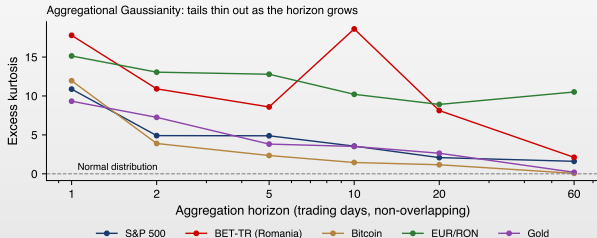
- ▣ This links heavy tails (Fact 2) to volatility clustering (Fact 6); a mixture drawn independently every day has heavy tails but no clustering (Seminar A6, A8)

## Fact 5: Intermittency in the Data



- Bursts of high volatility separated by calm periods; constant-volatility models cannot produce them
- 63-day rolling volatility  $\sqrt{P} \times$  the standard deviation over the window, log scale; common spikes: 2008–2009, March 2020
- EUR/RON volatility collapsed after 2012: an exchange-rate regime managed by the central bank

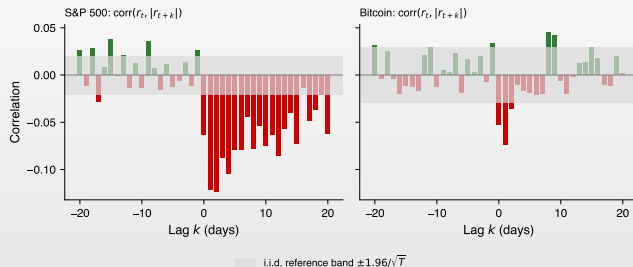
## Fact 4: Aggregational Gaussianity



- $h$ -day log return  $r_t(h) = r_t + \dots + r_{t-h+1}$  in non-overlapping blocks; the CLT pulls sums towards the Normal distribution, slowly because daily returns are dependent
- Excess kurtosis: S&P 500  $10.9 \rightarrow 1.6$  at  $h = 60$ , Bitcoin  $12.0 \rightarrow 0.1$
- At  $h = 60$  only 154 and 73 returns remain: reference SE  $\sqrt{24/n} = 0.39$  and  $0.57$  under i.i.d. normality, so these values are very imprecise
- $h = 60$  is a quarter for the S&P 500 but two months for Bitcoin

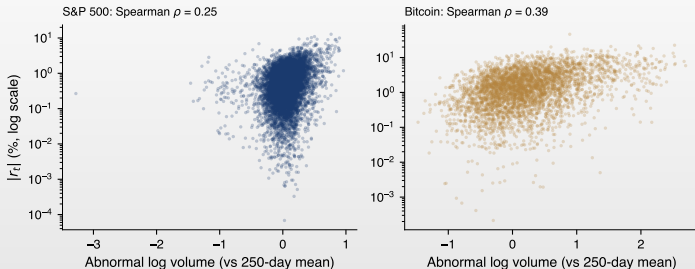
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## Fact 9: The Leverage Effect

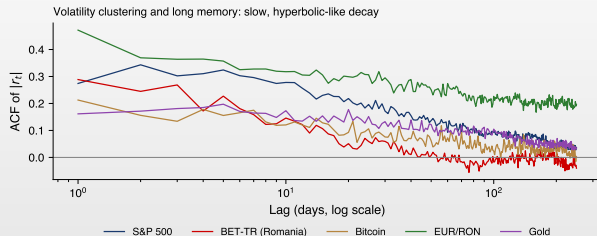


- $L(k) = \text{corr}(r_t, |r_{t+k}|)$ ,  $k > 0$ : today's return and a later size; S&P 500:  $L(1) = -0.12$ , still  $-0.07$  at  $k = 10$ ; no effect for  $k < 0$
- BET-TR  $-0.12$ ; weak for Bitcoin ( $-0.07$ ), negligible for gold ( $-0.02$ ), reversed for EUR/RON ( $+0.10$ )
- Possible explanations: financial leverage of firms [Christie \(1982\)](#), the retarded volatility model [Bouchaud, Matacz & Potters \(2001\)](#); models: GJR-GARCH, EGARCH ([Chapter 5](#))

## Fact 10: Volume and Volatility Move Together



- Abnormal log volume ( $\ln V_t$  minus its 250-day moving average) against  $|r_t|$ ; Spearman rank correlation 0.25 (S&P 500) and 0.39 (Bitcoin)
- Mixture-of-distributions hypothesis [Clark \(1973\)](#): information flow drives both price moves and volume
- Bitcoin volume is exchange-reported and partly **wash trading** (an exchange trading with itself) [Cong, Li, Tang & Yang \(2023\)](#)

Fact 8: Slow Decay of the ACF of  $|r_t|$ 

- **Short memory:**  $\sum_k |\rho_k| < \infty$ , e.g. exponential decay  $\rho_k \approx c \phi^k$  (GARCH(1,1), Chapter 5)
- **Long memory:** hyperbolic decay  $\rho_k \approx c k^{2d-1}$ ,  $0 < d < \frac{1}{2}$  Ding, Granger & Engle (1993)
- S&P 500: 0.27 at lag 1, 0.22 at lag 20, still 0.09 at lag 100; BET-TR decays fastest
- Slow empirical decay is not yet established long memory: breaks can imitate it (two slides on)

## Estimating the Memory Parameter $d$

- Long memory: spectral density  $f(\lambda) \approx G\lambda^{-2d}$  as  $\lambda \rightarrow 0$ ; semiparametric estimators use only the  $m$  lowest Fourier frequencies
- Local Whittle Robinson (1995):  $\sqrt{m}(\hat{d} - d) \rightarrow N(0, \frac{1}{4})$ ; log-periodogram regression Geweke & Porter-Hudak (1983):  $SE = \pi/\sqrt{24m}$ ; formulas in the [Appendix](#)

| $ r_t , m = T^{0.65}$          | S&P 500     | BET-TR      | Bitcoin     | EUR/RON     | Gold        |
|--------------------------------|-------------|-------------|-------------|-------------|-------------|
| $m$                            | 378         | 181         | 232         | 265         | 386         |
| Local Whittle $\hat{d}$ (SE)   | 0.48 (0.03) | 0.37 (0.04) | 0.35 (0.03) | 0.39 (0.03) | 0.40 (0.03) |
| Log-periodogram $\hat{d}$ (SE) | 0.50 (0.03) | 0.41 (0.05) | 0.35 (0.04) | 0.38 (0.04) | 0.42 (0.03) |

- $\hat{d} \approx 0.35$ – $0.50$  everywhere, at least 8 SEs from  $d = 0$  (short memory) with both estimators
- The S&P 500 is close to the stationarity boundary  $\frac{1}{2}$

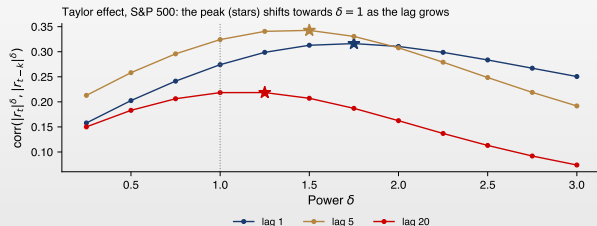
## Long Memory or Structural Breaks?

- Rare shifts in the level of volatility produce a slowly decaying ACF of  $|r_t|$  and  $\hat{d} > 0$  without long memory Diebold & Inoue (2001), Mikosch & Stărică (2004)
- Check: locate one variance break with the cumulative sum of squares Inclán & Tiao (1994), then re-estimate  $d$  (local Whittle) on each side

| $ r_t $ | Break date  | Volatility before / after (%) | $\hat{d}$ before (SE) | $\hat{d}$ after (SE) |
|---------|-------------|-------------------------------|-----------------------|----------------------|
| S&P 500 | 20 Jul 1998 | 12.6 / 19.3                   | 0.28 (0.04)           | 0.51 (0.03)          |
| BET-TR  | 17 Dec 2018 | 12.5 / 16.9                   | 0.19 (0.05)           | 0.37 (0.04)          |
| Bitcoin | 19 Jun 2022 | 74.9 / 47.7                   | 0.39 (0.04)           | 0.15 (0.05)          |
| EUR/RON | 27 Jun 2013 | 6.6 / 2.2                     | 0.35 (0.04)           | 0.35 (0.04)          |
| Gold    | 29 Nov 2005 | 13.2 / 18.3                   | 0.30 (0.03)           | 0.37 (0.03)          |

- $\hat{d}$  stays well above 0 in every sub-sample: one break does not explain the memory
- But  $\hat{d}$  moves (Bitcoin  $0.39 \rightarrow 0.15$ ): several breaks or a time-varying  $d$  remain open; the break dates are descriptive

## The Taylor Effect



- $\text{corr}(|r_t|^\delta, |r_{t-k}|^\delta)$  as a function of the power  $\delta > 0$ :  $\delta = 1$  absolute,  $\delta = 2$  squared returns
- S&P 500: maximum at  $\delta = 1.75$  (lag 1), 1.5 (lag 5) and 1.25 (lag 20): the optimum moves towards 1 at longer lags and is never at  $\delta = 2$
- Documented by [Taylor \(1986\)](#), named by [Granger & Ding \(1995\)](#); motivates power-GARCH models and absolute-value volatility measures

## Moments, Tail Index and Valid Inference

| Statistic                                          | Needs                                                 | With Hill $\hat{\alpha} \approx 2.5-3.5$ |
|----------------------------------------------------|-------------------------------------------------------|------------------------------------------|
| Mean, variance of $r_t$                            | $\alpha > 2$                                          | finite, slow convergence                 |
| ACF of $ r_t $ , robust ACF bands, $\tilde{Q}$     | $\alpha > 4$ for Normal limits                        | doubtful                                 |
| Kurtosis, ACF of $r_t^2$ , ARCH-LM, $Q$ on $r_t^2$ | $\alpha > 4$ to exist, $\alpha > 8$ for Normal limits | no standard inference                    |
| Skewness                                           | $\alpha > 6$ for a Normal limit                       | no standard inference                    |
| Quantiles, tail counts, Hill $\hat{\alpha}$        | no moments                                            | valid, with dependence-robust SEs        |

- With an infinite fourth moment the sample ACF of  $r_t^2$  converges slowly, to a limit that is not the Normal distribution [Davis & Mikosch \(1998\)](#): the  $\pm 1.96/\sqrt{T}$  band and the p-values of  $Q$  on  $r_t^2$  and ARCH-LM are not reliable
- The conclusion “volatility clusters” is robust; the exact magnitudes are not; one reason behind the Taylor effect:  $|r_t|$  needs fewer moments than  $r_t^2$

## Facts 7 and 11: Conditional Tails and Time Scales

### Fact 7: conditional heavy tails

- Fit a volatility model, e.g. GARCH (Chapter 5), and standardise:  $\hat{z}_t = r_t / \hat{\sigma}_t$
- The kurtosis of  $\hat{z}_t$  is lower than that of  $r_t$ , still above 3
- Models need heavy-tailed innovations (Student- $t$ , possibly skewed)

- Both facts are documented in Cont (2001); returns standardised by realised volatility from intraday data are close to the Normal distribution Andersen, Bollerslev, Diebold & Ebens (2001) (Chapter 9)

### Fact 11: asymmetry in time scales

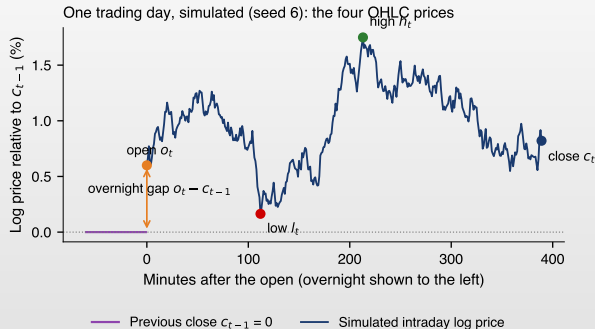
- Weekly volatility predicts daily volatility better than the reverse
- Traders with different horizons react at different frequencies
- **HAR-RV** model (Chapter 9)

## Where the Five Markets Differ

| Market  | Institutional feature                  | What differs                                                                                                                     | Candidate explanation                                                                            |
|---------|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Bitcoin | trades 24/7; $P = 365$                 | volatility 66.6% vs 18.0%; worst day $-46.5\%$ ; weak leverage ( $-0.07$ ); ACF of $ r_t $ at lag 100: 0.02 vs 0.09              | no corporate leverage; volatility rises after large moves of either sign                         |
| EUR/RON | BNR daily fixing, managed float        | volatility 4.4% but excess kurtosis 15.1; $\hat{\rho}_1 = +0.17$ ; largest moves October 2008 ( $+2.9\%$ , $-2.5\%$ )            | adjustments smoothed over several days                                                           |
| BET-TR  | total return, frontier/emerging market | skewness $-1.40$ ; worst day $-11.9\%$ on 19 Dec 2018 (bank-asset tax announced on 18 Dec, GEO 114/2018); $\hat{\rho}_1 = +0.06$ | policy shocks; non-synchronous trading of constituents<br>Campbell, Lo & MacKinlay (1997), ch. 3 |

- The explanations are hypotheses, not identified causes; low volatility is not low tail risk
- Heavy tails and volatility clustering hold in all five; autocorrelation and asymmetry depend on institutions

## Why Use Open, High, Low and Close?



- **OHLC**: the Open, High, Low and Close of a day;  $o_t, h_t, l_t, c_t$  their logs
- **Close-to-close**: the standard deviation of daily close-to-close log returns; one number per day
- **Range-based** estimators use the whole path: the range  $h_t - l_t$  is large on volatile days even if the close equals the open
- **Efficiency**: the ratio of the variances of two estimators

## Parkinson and Garman–Klass Estimators

- ▣ **Parkinson**, [Parkinson \(1980\)](#): uses only the daily range

$$\hat{\sigma}_P^2 = \frac{1}{4 \ln 2} \frac{1}{n} \sum_{t=1}^n (h_t - l_t)^2$$

- ▣ **Garman–Klass**, [Garman & Klass \(1980\)](#): combines the range with the open-to-close move

$$\hat{\sigma}_{GK}^2 = \frac{1}{n} \sum_{t=1}^n \left[ \frac{1}{2} (h_t - l_t)^2 - (2 \ln 2 - 1) (c_t - o_t)^2 \right]$$

- ▣ Derived for a Brownian motion (the continuous-time random walk, [Chapter 11](#)) without drift and without overnight jumps
  - ▶ efficiency relative to close-to-close: about 5 (Parkinson, Seminar A5) and 7–8 (Garman–Klass)

## Rogers–Satchell and Yang–Zhang Estimators

- **Rogers–Satchell**, [Rogers & Satchell \(1991\)](#): unbiased under a non-zero **drift** (expected trend of the log price)

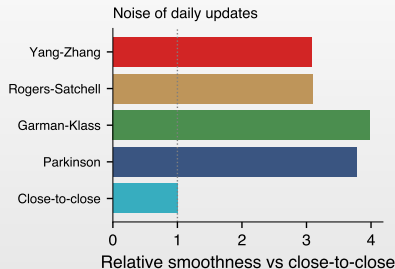
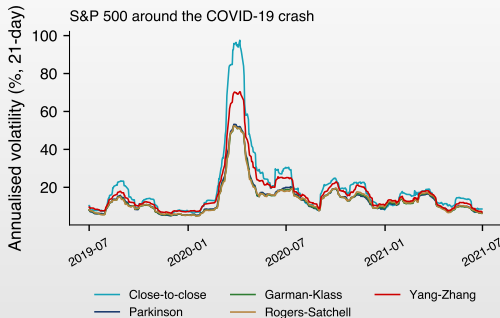
$$\hat{\sigma}_{RS}^2 = \frac{1}{n} \sum_{t=1}^n [(h_t - o_t)(h_t - c_t) + (l_t - o_t)(l_t - c_t)]$$

- **Yang–Zhang**, [Yang & Zhang \(2000\)](#): adds the overnight (close-to-open) jump

$$\hat{\sigma}_{YZ}^2 = \hat{\sigma}_{\text{overnight}}^2 + k \hat{\sigma}_{\text{open-close}}^2 + (1 - k) \hat{\sigma}_{RS}^2, \quad k = \frac{0.34}{1.34 + \frac{n+1}{n-1}}$$

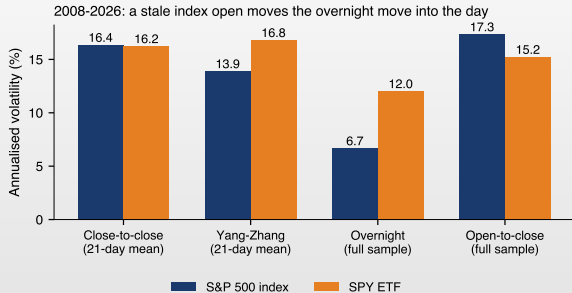
- ▶  $\hat{\sigma}_{\text{overnight}}^2$ : variance of  $o_t - c_{t-1}$ ;  $\hat{\sigma}_{\text{open-close}}^2$ : variance of  $c_t - o_t$
- ▶ minimum-variance combination among drift-independent estimators robust to opening jumps
- Range estimators are the bridge to realised volatility from intraday data ([Chapter 9](#))

## Estimators on the S&P 500



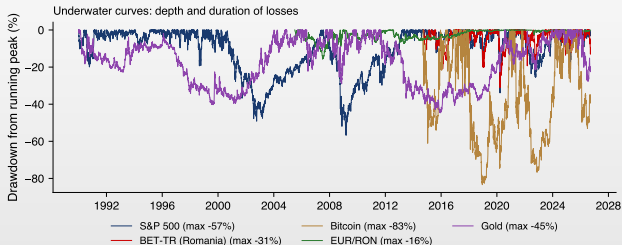
- 21-day windows, 2008–2026; averages: close-to-close 16.4%, Yang-Zhang 13.9%, Parkinson 13.0%, Garman-Klass 12.4%, Rogers-Satchell 12.3%
- Pure range estimators are lower: they miss overnight gaps; the range-based series are 3–4 times smoother, which is a property of the estimates, not by itself proof of efficiency

## Stale Opens: the S&P 500 Index versus SPY



- SPY (the ETF that tracks the S&P 500) opens at a traded price; the index open is computed from the constituents' first prices; at 9:30 many stocks have not traded yet, so their previous close enters the open
- Open = previous close: about 96% of days until 2005, 3.5% over 2008–2026 (up to 26% in 2012–2013)
- A stale open moves part of the overnight move into the day: index overnight volatility 6.7% vs 12.0% for SPY, whose open is a traded price
- Yang–Zhang falls below close-to-close on the index only; use a traded instrument for estimators that use the open

## Drawdowns Across Markets



- Drawdown  $DD_t = P_t / \max_{s \leq t} P_s - 1$ ;  
maximum drawdown  $MDD = \min_t DD_t$   
(Chapter 0)
- MDD: Bitcoin -83%, S&P 500 -57%, gold -45%, BET-TR -31% (March 2020), EUR/RON -16%
- S&P 500: 9 October 2007 (1,565.15) to 9 March 2009 (676.53): -56.8% needs +131% to recover
- The MDD grows with the sample length:  
compare only equal horizons (Appendix)

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## How Long Does Recovery Take?

|         | MDD  | Peak     | Trough   | Peak regained        |
|---------|------|----------|----------|----------------------|
| S&P 500 | −57% | Oct 2007 | Mar 2009 | Mar 2013 (5.5 years) |
| BET-TR  | −31% | Jan 2020 | Mar 2020 | Dec 2020 (0.9 years) |
| Bitcoin | −83% | Dec 2017 | Dec 2018 | Nov 2020 (3.0 years) |
| EUR/RON | −16% | Dec 2005 | Jul 2007 | Jan 2008 (2.0 years) |
| Gold    | −45% | Sep 2011 | Dec 2015 | Jul 2020 (8.9 years) |

- **Recovery duration:** the time from the peak until the price regains it
- Depth and duration are different risks: gold's drawdown was shallower than Bitcoin's but lasted three times longer
- Fast V-shaped recoveries (BET-TR 2020) versus long underwater periods (gold 2011–2020)
- For EUR/RON a “drawdown” is a fall of the euro against the leu

## Sharpe, Sortino and Calmar Ratios

- **Sharpe ratio** Sharpe (1966), Sharpe (1994):  $SR = (\mu_{\text{ann}} - r_f) / \sigma_{\text{ann}}$  on **simple** returns,  $\mu_{\text{ann}} = P\bar{R}$ ;  $r_f = 0$  here, for comparability
  - ▶ S&P 500 1990–2026:  $\mu_{\text{ann}} = 9.95\%$  (mean log return 8.33% plus  $\approx \sigma^2/2 = 1.62$  pp),  $\sigma_{\text{ann}} = 17.97\%$ ,  $SR = 0.55$
- **Sortino ratio** Sortino & van der Meer (1991):  $SoR = (\mu_{\text{ann}} - r_f) / \sigma_{\text{ann}}^-$ ,  $\sigma^- = \sqrt{\frac{1}{T} \sum_{t=1}^T \min(R_t, 0)^2}$ 
  - ▶ the average runs over all  $T$  days, positive days counted as 0; S&P 500:  $\sigma_{\text{ann}}^- = 12.68\%$ ,  $SoR = 0.78$
- **Calmar ratio** =  $\text{CAGR} / |MDD|$ ; S&P 500:  $8.69\% / 56.78\% = 0.15$ , driven by one episode

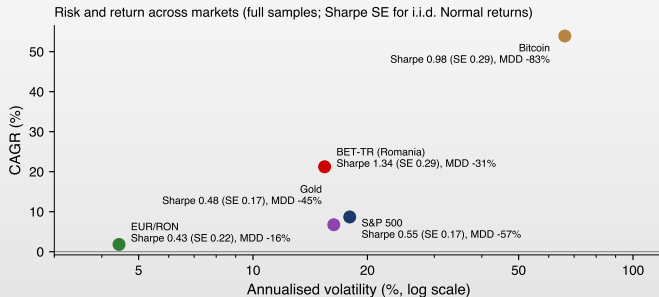
## Sharpe Ratios with Standard Errors

- Delta method, i.i.d. returns from the Normal distribution [Lo \(2002\)](#):  $\text{Var}(\widehat{SR}_d) \approx (1 + SR_d^2/2)/T$ ; annualised  $SE(\widehat{SR}_{\text{ann}}) \approx \sqrt{P(1 + SR_d^2/2)/T} \approx 1/\sqrt{Y}$ : the span matters, not the frequency

|                           | S&P 500     | BET-TR      | Bitcoin     | EUR/RON     | Gold        |
|---------------------------|-------------|-------------|-------------|-------------|-------------|
| Years Y                   | 36.7        | 12.0        | 12.0        | 21.2        | 36.7        |
| Sharpe ratio (SE, Normal) | 0.55 (0.17) | 1.34 (0.29) | 0.98 (0.29) | 0.43 (0.22) | 0.48 (0.17) |

- Clustering: HAC or studentised block-bootstrap tests [Ledoit & Wolf \(2008\)](#); they still need finite fourth moments, which the Hill estimates put in doubt
  - ▶ BET-TR vs S&P 500, common days 2014–2026: difference 0.61,  $z = 1.75$  ( $p = 0.08$ ), block-bootstrap 95% CI  $[-0.01, 1.21]$  (Seminar B13)
- Calmar has no usable sampling theory: its drawdown grows with the sample length

## Risk and Return Across Markets



- Sharpe ratios on simple returns, SEs (i.i.d. Normal) in the labels; full table in the [Appendix](#)
- Different sample periods: not a horse race
- S&P 500 without dividends; BET-TR with dividends and in RON, where short-term rates were well above USD rates: with  $r_f = 0$  its Sharpe ratio is inflated
- Heavy tails and negative skewness flatter the Sharpe ratio; report it with drawdowns and tail statistics

## Case Study (1/5): Questions and Target

- ▣ Paper: Guyon & Lekeufack (2023), *Quantitative Finance*
- ▣ Questions
  - ▶ how much of an equity index's volatility can its past daily returns explain?
  - ▶ what function describes this relationship?
  - ▶ target: the 30-day implied volatility index (VIX for the S&P 500) and the 9-day VIX; next-day realised volatility from 5-minute returns (Sections 3.4–3.5 of the paper)
  - ▶ one formula must produce clustering, slow decay and leverage (Facts 6, 8, 9) together; builds on Bollerslev, Hood, Huss & Pedersen (2018)
- ▣ Why it matters: if volatility is mostly a function of the price path, it is mostly endogenous; a **path-dependent volatility (PDV)** model needs little extra randomness, unlike stochastic volatility models (Chapters 11–12)
- ▣ Data: daily closes of the S&P 500, Euro Stoxx 50, FTSE 100, DAX and Nikkei 225; training 1 Jan 2000 – 31 Dec 2018, test 1 Jan 2019 – 15 May 2022

## Case Study (2/5): Model and Features

- Model, eq. (3.7) in the paper, with simple daily returns  $r_{t_i} = S_{t_i}/S_{t_{i-1}} - 1$

$$\sigma_t = \beta_0 + \beta_1 R_{1,t} + \beta_2 \Sigma_t, \quad R_{1,t} = \sum_{t_i \leq t} K_1(t - t_i) r_{t_i}, \quad \Sigma_t = \left( \sum_{t_i \leq t} K_2(t - t_i) r_{t_i}^2 \right)^{1/2}$$

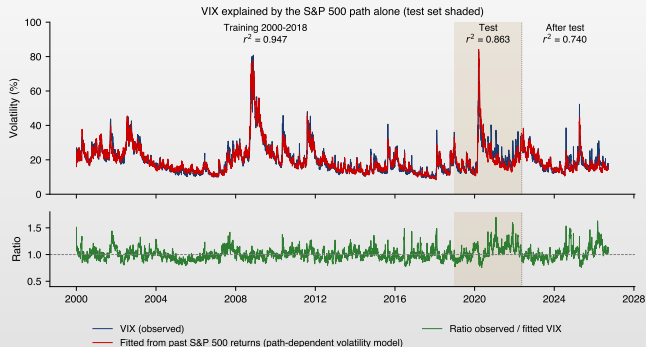
- ▶  $\sigma_t$ : implied volatility at the close of day  $t$ ;  $\beta_1 < 0$  gives the leverage effect,  $\beta_2 \in (0, 1)$  gives clustering, as in GARCH
- Kernels: **time-shifted power law (TSPL)**, eq. (3.9) in the paper:  $K(\tau) \propto (\tau + \delta)^{-\alpha}$ ,  $\alpha > 1$ ,  $\delta > 0$ , on 1,000 past days
  - ▶ large weight on the last few days (short memory) and a power-law tail (long memory, Fact 8)
- Seven parameters by nonlinear least squares on the training set; scores: RMSE (root mean squared error) and  $r^2 = 1 - \sum(\sigma_t - \hat{\sigma}_t)^2 / \sum(\sigma_t - \bar{\sigma})^2$ ; estimation details: [▶ Appendix](#)

## Case Study (3/5): Published and Replicated Scores

| Index                          | Implied volatility (paper, Table 2) |            | Realised volatility (paper, Table 4) |            |
|--------------------------------|-------------------------------------|------------|--------------------------------------|------------|
|                                | $r^2$ train                         | $r^2$ test | $r^2$ train                          | $r^2$ test |
| S&P 500 (VIX)                  | 0.946                               | 0.855      | 0.738                                | 0.654      |
| S&P 500 (VIX), our replication | 0.947                               | 0.863      | —                                    | —          |
| S&P 500 (9-day VIX)            | 0.876                               | 0.914      | —                                    | —          |
| Euro Stoxx 50                  | 0.929                               | 0.913      | 0.672                                | 0.682      |
| Nikkei 225                     | 0.890                               | 0.800      | 0.563                                | 0.504      |

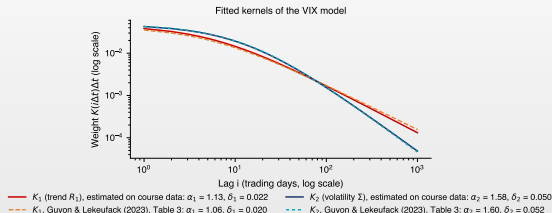
- Past index returns alone explain 80–92% of the variance of implied volatility out of sample; FTSE 100 and DAX in the [Appendix](#)
- On our data (S&P 500 and VIX): RMSE 0.020 training and 0.034 test versus 0.020 and 0.035 in Table 2 of the paper
- Realised volatility scores are lower because 5-minute realised volatility is a noisy target (Remark 3 in the paper); this course uses no 5-minute data, so Table 4 of the paper is not replicated

## Case Study (4/5): Observed and Fitted VIX



- Fitted on 3 Jan 2000 – 31 Dec 2018 (4,779 days), started from Table 3 of the paper, then parameters frozen; test set shaded
- After the test set, 16 May 2022 – 18 Sep 2026:  $r^2 = 0.740$ , RMSE 0.026: a lower RMSE, but the VIX varied less (standard deviation 5.0 versus 9.3 points), so  $r^2$  falls
- Ratio observed/fitted: mean 0.85 in April–May 2020 and 1.16 from October 2020 to May 2022; 8 April 2025: VIX 52.3 vs 39.2 fitted

## Case Study (5/5): Fitted Kernels and Limits



- Daily weights of  $K_2$ , log-log scale:  $\hat{\alpha}_2 = 1.58$ ,  $\hat{\delta}_2 = 0.050$  versus 1.60, 0.052 in Table 3 of the paper
- The last 5 days carry 20% of the weight, the days beyond lag 250 still 10%: short and long memory in one kernel
- Limits stated by the authors: a small exogenous part remains, the ratio of observed to fitted VIX mean-reverts around 1 and needs a stochastic factor (Section 5 of the paper)
- Open extension: does the same formula describe the realised volatility of Bitcoin or of BET-TR? (project seed)

 MFM\_ch1\_path\_dependent\_vol

## From Evidence to Model Requirements

| Evidence (five markets)                                                            | Model requirement                                                    | Where in this course                   |
|------------------------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------|
| quantile kurtosis $> 0$ everywhere;<br>Hill $\hat{\alpha} \approx 2.5\text{--}3.5$ | heavy-tailed innovations, also<br>conditionally                      | Ch. 5, Ch. 7 (extreme value<br>theory) |
| ACF of $ r_t $ positive for months;<br>$\hat{d} \approx 0.35\text{--}0.50$         | persistent, time-varying volatility;<br>long memory or breaks        | Ch. 5, Ch. 9                           |
| $L(1) = -0.12$ (S&P 500,<br>BET-TR); reversed for EUR/RON                          | asymmetric response to bad news<br>where it exists                   | Ch. 5 (GJR-GARCH,<br>EGARCH)           |
| robust $\hat{\rho}_1$ economically small<br>except EUR/RON                         | approximately uncorrelated returns;<br>market microstructure matters | Ch. 2, Ch. 10                          |
| quantile skewness negative only for<br>the S&P 500                                 | asymmetry is market-specific                                         | Ch. 7–8                                |

- The random walk with i.i.d. increments from the Normal distribution (Bachelier–Black–Scholes world) violates Facts 2, 3 and 5–11; Fact 4 holds trivially
- ARCH-LM significance supports clustering descriptively; with doubtful fourth moments it is not a formal proof

## Open Question: Are Tail Indices Stable over Time?

- The tail index  $\alpha$  is usually treated as a constant; crypto-assets changed in structure: futures (2017), spot Bitcoin ETFs (January 2024), institutional holders
  - ▶ [Pele et al. \(2023\)](#): the tail factor separates cryptos from other asset classes
- A typical AI-assistant proposal: “institutional money since the spot ETFs has tamed Bitcoin’s tails”
- Discovery loop in one pass
  - ▶ **literature**: ask for peer-reviewed work with DOIs, then resolve each DOI yourself; [Hoga \(2017\)](#): change-point tests for the tail index valid under  $\beta$ -mixing (e.g. GARCH-type) dependence
  - ▶ **hypotheses**: a change in tail shape, a fall in volatility only (scale does not move  $\alpha$ ), sampling noise
  - ▶ **data and code**: replicate the chapter’s Hill estimates first (Bitcoin 4.2, S&P 500 3.1)
- Fixed in advance:  $k = 2.5\%$  of  $n$ , the break date 11 January 2024, block-bootstrap SEs, a placebo asset

## Mini-Case: Did Bitcoin's Tail Get Thinner after the ETFs?

- Test: Hill index of  $|r_t|$ ,  $k = 2.5\%$  of  $n$ ;  $H_0 : \alpha_{\text{post}} = \alpha_{\text{pre}}$  against  $H_1 : \alpha_{\text{post}} > \alpha_{\text{pre}}$ ,  
 $z = (\hat{\alpha}_{\text{post}} - \hat{\alpha}_{\text{pre}}) / \sqrt{s_{\text{pre}}^2 + s_{\text{post}}^2}$ ,  $s$  from a moving-block bootstrap (blocks of 20 days)
- Before (Sep 2014 – 10 Jan 2024,  $k = 85$ ):  $\hat{\alpha} = 4.16$  ( $s = 0.52$ ); after (11 Jan 2024 – Sep 2026,  $k = 24$ ):  
 $\hat{\alpha} = 4.31$  ( $s = 0.91$ );  $z = 0.15$
- Placebo, S&P 500 at the same date:  $2.82 \rightarrow 2.86$ ,  $z = 0.03$
- **Minimum detectable change**, one-sided 5% test with 80% power:  $(1.645 + 0.842) \times 1.05 = 2.6$ 
  - ▶ a normal approximation, rough with 24 tail observations: the data cannot distinguish “unchanged” from a large thinning of the tail
- “Not significant” is not “no effect”: wait for more data or pool assets

## Your Turn: A Project Seed

- ▣ **Question:** did the tail index of Bitcoin and Ethereum change with institutionalisation (futures 2017, spot ETFs 2024)?
  - ▶ raw and GARCH-standardised returns (shape versus scale); loss and gain tails separately
- ▣ **Replicate first:** the chapter's Hill table and the mini-case numbers
- ▣ **Pre-register:**  $H_0$ ,  $H_1$ , the rule for  $k$ , break dates and bootstrap, committed before the results
- ▣ **Report:** the change-point test of [Hoga \(2017\)](#) as a Quantlet, estimates, both SEs, the minimum detectable change
- ▣ **Log:** prompts in `AI_USE.md`, every caught error (an i.i.d. SE, a date chosen from the data) in `AI_ERRORS.md`; reusable prompts and checks in the [▶ Appendix](#)

## Key Takeaways and Next: Chapter 2

### Takeaways

- ▣ Returns are uncorrelated but not independent: their size is predictable
- ▣ Heavy tails are the rule: tail indices near 3, so the fourth moment very likely does not exist and kurtosis-based inference fails
- ▣ Volatility is persistent (long memory or breaks) and reacts asymmetrically to losses in equity markets
- ▣ Every test needs a reference distribution that the data may violate

### Check yourself

- ▣ Why does the  $Q(10)$  on Bitcoin returns have  $p = 0.028$  but the robust  $\tilde{Q}(10)$   $p = 0.29$ ?
- ▣ Why can the Hill index be estimated when the kurtosis does not exist?
- ▣ **Chapter 2:** are returns predictable at all?

## Appendix: Formula Sheet with Assumptions

### Returns and annualisation

- ▣  $R_t = P_t/P_{t-1} - 1$ ,  $r_t = \ln(1 + R_t)$ ,  $r_t(k) = \sum_{j=0}^{k-1} r_{t-j}$ ,  
 $R_{p,t} = \sum_i w_i R_{i,t}$
- ▣  $\text{CAGR} = (P_T/P_0)^{1/Y} - 1$
- ▣  $\sigma_{\text{ann}} = \sqrt{P} \hat{\sigma}$ : uncorrelated returns, constant variance

### Tests

- ▣  $JB = T[\hat{S}^2/6 + (\hat{K} - 3)^2/24]$ :  $\chi_2^2$  under i.i.d. Normal
- ▣  $Q(m) = T(T+2) \sum_{k=1}^m \hat{\rho}_k^2 / (T-k)$ :  $\chi_m^2$  under i.i.d.
- ▣  $\tilde{Q}(m) = \sum_k \hat{\rho}_k^2 / \hat{\tau}_k$ :  $\mathbb{E}r^4 < \infty$ , zero cross terms
- ▣ Hill  $\text{SE}(\hat{\alpha}) \approx \hat{\alpha} / \sqrt{k}$ : independent Pareto tail

### Volatility

- ▣ Parkinson  $\frac{1}{4 \ln 2} (\overline{h} - l)^2$ : driftless, no overnight jumps
- ▣ Mixture  $K = 3 \mathbb{E}[\sigma^4] / \mathbb{E}[\sigma^2]^2$ :  $\sigma_t$  independent of  $\varepsilon_t$

### Performance

- ▣  $SR = (\mu - r_f) / \sigma$  on simple returns;  $\text{SE} \approx 1/\sqrt{Y}$  for i.i.d. Normal returns
- ▣  $SoR = (\mu - r_f) / \sigma^-$
- ▣  $DD_t = P_t / \max_{s \leq t} P_s - 1$ ; Calmar =  $\text{CAGR} / |MDD|$

## Appendix: Prices Before Computers, the Stock Ticker



Stock ticker designed by Thomas Edison  
(Smithsonian collection)

Photo: Jaclyn Nash, Smithsonian National Museum of  
American History; public domain; Wikimedia Commons

- Before computers, trades reached brokers as prices printed on a paper **ticker tape**
- A daily close is one snapshot of this stream of prices; returns are built from such snapshots
- The choice of snapshot (close, fixing, a midnight snapshot) matters, as [Chapter 0](#) showed
- Prices at the level of individual trades: market microstructure, [Chapter 10](#)

## Appendix: The People Behind the Stylised Facts (1/2)



Carl Friedrich Gauss (1777–1855)

Painting: C.A. Jensen (1840); public domain;  
[Wikimedia Commons](#)

- The Normal distribution: the benchmark that returns violate



Louis Bachelier (1870–1946)

Photo: unknown author; public domain;  
[Wikimedia Commons](#)

- Prices as a random walk, in his thesis [Bachelier \(1900\)](#)

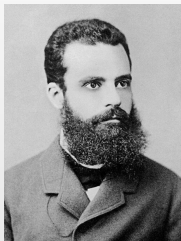


William S. Gosset, "Student" (1876–1937)

Photo: unknown author; public domain;  
[Wikimedia Commons](#)

- The Student-t distribution [Student \(1908\)](#), today a model for heavy tails

## Appendix: The People Behind the Stylised Facts (2/2)



Vilfredo Pareto (1848–1923)

Photo: unknown author (1870s); public domain; [Wikimedia Commons](#)



Benoît Mandelbrot (1924–2010)

Photo: Rama (2007); CC BY-SA 2.0 France; [Wikimedia Commons](#)

- Pareto: **power laws** for large incomes, the ancestor of heavy-tail models
- Mandelbrot: cotton prices have heavy tails and the Normal-distribution model fails [Mandelbrot \(1963\)](#)
- Their tail index  $\alpha$  is what the Hill estimator in this chapter measures



## Appendix: Do Cont's Facts Still Hold? Recent Evidence

- Ratliff-Crain, Van Oort, Koehler & Tivnan (2025)
  - ▶ re-test all 11 facts on intraday returns of the Dow 30 stocks, October 2018 – March 2019
  - ▶ conclusive evidence for 8 facts, no support for the remaining 3
- Heavy tails beyond the Student- $t$ : estimation of alpha-stable distributions, whose tail index is below 2 [Pele \(2014\)](#)
- Entropy-based measures of tail risk as an alternative to kurtosis [Pele, Lazar & Dufour \(2017\)](#)
- Crypto-assets: the same facts, with a tail factor that separates them from other asset classes [Pele et al. \(2023\)](#)

## Appendix: Annual Variance under Autocorrelation

- Weakly stationary returns (constant mean and variance, autocovariances depending only on the lag):

$$\text{Var}\left(\sum_{j=1}^h r_{t+j}\right) = h\sigma^2 \left[1 + 2 \sum_{j=1}^{h-1} (1 - j/h) \rho_j\right]$$

- Autoregression of order 1 (AR(1)),  $\rho_j = \rho^j$ : the factor  $\sqrt{VR(h)}$ , with  $VR(h)$  the variance ratio  $\text{Var}(h\text{-day sum})/(h\sigma^2)$ :  $\sqrt{VR(h)} \rightarrow \sqrt{(1+\rho)/(1-\rho)}$  for large  $h$ 
  - ▶ EUR/RON,  $\rho_1 = 0.17$ : factor 1.19, annual volatility 4.4%  $\rightarrow$  5.3%
  - ▶ S&P 500,  $\rho_1 = -0.079$ : factor 0.92
- Using the sample lag-1 autocorrelation as an AR(1) parameter is an assumption: check the whole ACF (Seminar A2); variance-ratio tests in [Chapter 2](#)

## Appendix: What JB Tests, and Tests for Time Series

- JB is a Lagrange Multiplier test [Jarque & Bera \(1980\)](#): under i.i.d. returns from the Normal distribution  $\sqrt{T}(\hat{S}, \hat{K} - 3) \rightarrow N(0, \text{diag}(6, 24))$ , hence  $\chi_2^2$
- GARCH-type dependence changes  $\text{Var}(\hat{S})$  and  $\text{Var}(\hat{K})$ : a rejection mixes the marginal distribution and independence
- [Bai & Ng \(2005\)](#): tests of skewness, kurtosis and normality of the marginal distribution of a time series, studentised with HAC variances
  - ▶ their separate skewness and kurtosis tests (null  $S = 0$  or  $K = 3$  without normality) need  $\mathbb{E}r^6 < \infty$  and  $\mathbb{E}r^8 < \infty$

## Appendix: The Robust Portmanteau Test and Its Assumptions

- ▣  $\tilde{Q}(m) = \sum_{k=1}^m \hat{\rho}_k^2 / \hat{\tau}_k \sim \chi_m^2$  Lobato, Nankervis & Savin (2001) needs  $\mathbb{E}r^4 < \infty$  and zero cross terms  $\mathbb{E}[x_t^2 x_{t-k} x_{t-l}] = 0$  for  $k \neq l$
- ▣ Dalla, Giraitis & Phillips (2022) keep only the significant cross terms in the covariance of the  $\hat{\rho}_k$ : none for the S&P 500, BET-TR and Bitcoin
- ▣  $\hat{\tau}_k$  is the variance under  $\rho_k = 0$ ; to compare a non-zero  $\rho_1$  across periods use the HAC SE of the slope of  $r_t$  on  $r_{t-1}$  (Seminar B6)
- ▣ ACF of  $|r_t|$  and of  $r_t^2$ : Normal limits need  $\alpha > 4$  and  $\alpha > 8$  Davis & Mikosch (1998)

## Appendix: Hill Estimator, Asymptotics and Alternatives

- ▣ Hill = maximum likelihood estimator of a Pareto tail above  $u = X_{(k+1)}$ , conditional on  $u$  (Seminar A11)
- ▣ If  $k \rightarrow \infty$  and  $k/n \rightarrow 0$ , and the tail is Pareto up to a second-order term

$$\sqrt{k}(\hat{\alpha}_k - \alpha) \rightarrow N(0, \alpha^2)$$

- ▶ if  $k$  grows too fast, a bias term appears: the tail is only approximately Pareto
- ▣ Dependent data (GARCH-type):  $\hat{\alpha}$  stays consistent, but extremes cluster and the variance exceeds  $\alpha^2/k$  [Hsing \(1991\)](#)
- ▣ Choosing  $k$ : plot  $\hat{\alpha}$  against  $\log k$  ("alt-Hill" plot) [Drees, Resnick & de Haan \(2000\)](#)
- ▣ Regression alternative: log rank on log size, with rank shifted by  $-\frac{1}{2}$ ;  $SE = \hat{\alpha}\sqrt{2/k}$  [Gabaix & Ibragimov \(2011\)](#)

## Appendix: Two Tails and the Tail of $|r_t|$

- $P(|X| > x) = P(X > x) + P(X < -x)$ : with indices  $\alpha_-$  (losses) and  $\alpha_+$  (gains),  $|X|$  has index  $\min(\alpha_-, \alpha_+)$
- At the same  $k$  the estimates are not comparable: the  $k$  largest  $|r_t|$  come from both tails; common threshold  $u$ : the lower of the two one-sided thresholds at  $k = 2.5\%$  of  $n$

|         | $u$ (%) | $k$ for $ r_t $ | $\hat{\alpha},  r_t $ | $\hat{\alpha}_-, \text{losses}$ | $\hat{\alpha}_+, \text{gains}$ |
|---------|---------|-----------------|-----------------------|---------------------------------|--------------------------------|
| S&P 500 | 2.17    | 508             | 2.90                  | 2.85                            | 2.96                           |
| BET-TR  | 1.79    | 153             | 2.54                  | 2.20                            | 3.06                           |
| Bitcoin | 7.11    | 233             | 3.02                  | 2.71                            | 3.36                           |
| EUR/RON | 0.54    | 295             | 2.28                  | 2.63                            | 2.04                           |
| Gold    | 1.98    | 513             | 3.11                  | 2.97                            | 3.28                           |

- At a common  $u$ , the  $|r_t|$  estimate lies between the two tails; Bitcoin's 4.2 at  $k = 2.5\%$  reflects second-order bias: report both one-sided indices

## Appendix: Estimating $d$ , the Formulas

- ▣ Fourier frequencies  $\lambda_j = 2\pi j/T$ , periodogram  $I(\lambda_j)$  of the demeaned series;  $m$  lowest frequencies, here  $m = T^{0.65}$
- ▣ Log-periodogram regression [Geweke & Porter-Hudak \(1983\)](#): regress  $\ln I(\lambda_j)$  on  $-2 \ln(2 \sin(\lambda_j/2))$ ; the slope is  $\hat{d}$ ,  $SE = \pi/\sqrt{24m}$
- ▣ Local Whittle [Robinson \(1995\)](#):

$$\hat{d} = \arg \min_d \left[ \ln \left( \frac{1}{m} \sum_{j=1}^m \lambda_j^{2d} I(\lambda_j) \right) - \frac{2d}{m} \sum_{j=1}^m \ln \lambda_j \right], \quad \sqrt{m}(\hat{d} - d) \rightarrow N(0, \frac{1}{4})$$

- ▣ Local Whittle is robust to non-Gaussianity; a larger  $m$  lowers the SE but lets short-run (GARCH) dynamics bias  $\hat{d}$  (Seminar B15)

## Appendix: Markets in Detail

- ▣ **Crypto-assets:** an asset class with its own statistical profile [Härdle, Harvey & Reule \(2020\)](#); tail, memory and moment factors classify cryptos as a separate asset class, mainly through the tail factor [Pele et al. \(2023\)](#); crypto modelling and the CRIX (CRypto IndeX) [Härdle, Chen & Overbeck \(2017\)](#); Chapter 16
- ▣ **EUR/RON:** BNR reference rate from 1 July 2005, the new leu (RON, 1 RON = 10,000 old lei); long calm periods interrupted by sudden adjustments; EUR/RON market (interbank) quote series showed spurious negative autocorrelation (quote errors)
- ▣ **BET-TR:** the bank-asset tax announced on 18 December 2018 was adopted on 28 December 2018 as Government Emergency Ordinance (GEO) 114/2018; policy shocks produce single-day crashes without comparable rallies
- ▣ **Non-synchronous trading:** index constituents do not all trade at every moment, so the index reflects news with a delay, creating positive autocorrelation [Campbell, Lo & MacKinlay \(1997\)](#), ch. 3

## Appendix: The Maximum Drawdown of a Brownian Motion

- Driftless Brownian motion of the log price with annual volatility  $\sigma$  over  $T$  years: the maximum **log** drawdown  $D = -\ln(1 + MDD) \geq 0$  has

$$\mathbb{E}[D] = \sqrt{\pi/2} \sigma \sqrt{T} \quad \text{Magdon-Ismail, Atiya, Pratap \& Abu-Mostafa (2004)}$$

- $MDD = e^{-D} - 1$ ; with a positive drift  $\mathbb{E}[D]$  grows only like  $\ln T$
- Gold 1990–2026:  $\sigma = 16.3\%$ ,  $T = 36.7$ : benchmark  $\mathbb{E}[D] = 1.24$  versus observed 0.592 (−44.7%) (Seminar A4)
- Drawdowns and Calmar ratios from samples of different length are not comparable

## Appendix: Sharpe Ratios beyond the Normal Distribution

- i.i.d., any distribution with  $\mathbb{E}r^4 < \infty$ :  $\text{Var}(\widehat{SR}_d) \approx (1 - S SR_d + \frac{K-1}{4} SR_d^2)/T$  [Opdyke \(2007\)](#); here the SEs change by at most 0.02
- The Deflated Sharpe Ratio corrects for non-normality and multiple testing ([Chapter 13](#))

|         | CAGR  | Sharpe | Sortino | Calmar |
|---------|-------|--------|---------|--------|
| S&P 500 | 8.7%  | 0.55   | 0.78    | 0.15   |
| BET-TR  | 21.2% | 1.34   | 1.87    | 0.68   |
| Bitcoin | 53.9% | 0.98   | 1.45    | 0.65   |
| EUR/RON | 1.8%  | 0.43   | 0.65    | 0.12   |
| Gold    | 6.7%  | 0.48   | 0.69    | 0.15   |

 MFM\_ch1\_risk\_measures

## Appendix: Case Study, Estimation Details

- Targets (Sections 3.4–3.5 of the paper): the 30-day implied volatility index of each index and the 9-day VIX; realised volatility of day  $t + 1$ ; volatility in decimals ( $0.03 = 3$  volatility points)
- Kernels, eq. (3.9) in the paper:  $K(\tau) = Z_{\alpha, \delta}^{-1}(\tau + \delta)^{-\alpha}$ , normalised so that  $\sum_i K(i\Delta t)\Delta t = 1$  on the 1,000-day window,  $\Delta t = 1/252$
- Estimation:  $(\beta_0, \beta_1, \beta_2, \alpha_1, \delta_1, \alpha_2, \delta_2)$  by nonlinear least squares on the training set, initialised using a lasso regression on features built with exponential kernels
- Our replication: started from Table 3 of the paper;  $\hat{\beta}_0 = 0.052$ ,  $\hat{\beta}_1 = -0.090$  ( $-0.095$ ),  $\hat{\beta}_2 = 0.868$  ( $0.82$ ),  $\hat{\alpha}_1 = 1.13$  ( $1.06$ ); parameter differences alone do not measure precision
- The 9-day VIX exists only from 2011 (shorter training set); the restricted models ( $\beta_1 = 0$ ,  $\beta_2 = 0$ ) are Seminar C3

## Appendix: Case Study, Complete Published Results

| Index               | Implied volatility (paper, Table 2) |            | Realised volatility (paper, Table 4) |            |
|---------------------|-------------------------------------|------------|--------------------------------------|------------|
|                     | $r^2$ train                         | $r^2$ test | $r^2$ train                          | $r^2$ test |
| S&P 500 (VIX)       | 0.946                               | 0.855      | 0.738                                | 0.654      |
| S&P 500 (9-day VIX) | 0.876                               | 0.914      | –                                    | –          |
| Euro Stoxx 50       | 0.929                               | 0.913      | 0.672                                | 0.682      |
| FTSE 100            | 0.925                               | 0.870      | 0.650                                | 0.617      |
| DAX                 | 0.934                               | 0.918      | 0.722                                | 0.557      |
| Nikkei 225          | 0.890                               | 0.800      | 0.563                                | 0.504      |

- VIX parameters:  $\beta_0 = 0.057$ ,  $\beta_1 = -0.095$ ,  $\beta_2 = 0.82$ ; across all targets  $\alpha_2 \in [1.2, 2.3]$  (Tables 3 and 5 of the paper): long memory near criticality (Section 3.2 of the paper)
- Dropping  $\Sigma_t$  hurts the most; the model beats two ARCH benchmarks and four published path-dependent models (Figures 3.8–3.9 and Appendix C of the paper)
- Its Markovian 4-factor version, driven by Brownian motion, gives estimated values of about 0.12–0.17 of the Hurst exponent  $H$  ( $H < 0.5$ : a path “rougher” than Brownian motion) for daily realised volatility, as in data (Section 4.6 and Table 9 of the paper): the roughness is only apparent at daily scales

## Appendix: The Discovery Loop, Reusable Prompts

- **Literature:** Find peer-reviewed work since 2015 on time variation or structural breaks in tail indices of asset returns, with tests valid under GARCH-type dependence. For each: DOI, estimator, test, data. Flag entries you are unsure of.
- **Hypothesis:** State competing hypotheses for a change in Bitcoin's tail after the 2024 spot ETFs. What does each predict for the Hill index of raw and of GARCH-standardised returns?
- **Data and code:** Use daily Bitcoin closing prices in USD; log returns on its own calendar; Hill index of  $|r|$  with  $k = 2.5\%$  of  $n$  before and after 11 January 2024; moving-block bootstrap standard errors.
- **Robustness:** Write a pre-registered plan:  $k$  from 1% to 5% of  $n$ , loss and gain tails, GARCH-standardised residuals, block lengths 10 to 50 days, a placebo date and a placebo asset.
- **Critique:** Referee this design. With 24 tail observations after the break, what is the power against a change of 1 in the tail index?

## Appendix: What the Researcher Must Check

- ❑ **Wrong return or price series:** reproduce the chapter's Hill table before anything else
- ❑ **Choice of  $k$  after seeing the result:** fix the rule ( $k = 2.5\%$  of  $n$ ) in advance; show the Hill plot for 1–5%
- ❑ **i.i.d. standard errors:**  $\hat{\alpha}/\sqrt{k}$  ignores volatility clustering; compare with a block bootstrap (for the S&P 500 before 2024 it is 2.2 times larger)
- ❑ **Break date chosen from the data:** use a date fixed by the event, or a sup-type change-point test with its own critical values [Hoga \(2017\)](#)
- ❑ **Bad ticks create fake tails:** list the ten largest  $|r_t|$  and match each to a dated event
- ❑ **Low power:** report the minimum detectable change: 2.8 SEs of the difference for a two-sided 5% test with 80% power, 2.5 for a one-sided test

## Appendix: Practical Checklist for Empirical Work

- ▣ **Know your data:** price vs total-return index; adjusted vs raw prices; fixing vs traded price; official reference rates where possible (EUR/RON)
- ▣ **Clean before you model:** non-trading days, holidays, stale quotes, obvious errors; match the largest moves to real events
- ▣ **Use log returns** for single-asset dynamics, simple returns for portfolios
- ▣ **Report robust statistics:** quantiles, tail counts and tail indices next to kurtosis
- ▣ **Annualise carefully:** 252 vs 365 periods; the  $\sqrt{P}$  rule needs uncorrelated returns with constant variance
- ▣ **Reproducibility:** code and data on GitHub, as Quantlets

## Appendix: Further Study on Quantinar

### Advanced courses on quantinar.com

- ▣  Statistics of Financial Markets
- ▣  Tukey's g- and h- transformations
- ▣  Applied Time Series Analysis with Python

### Code, data and essential reading

- ▣ All charts and tables: [github.com/danpele/MFM/tree/main/Quantlets/Ch\\_01](https://github.com/danpele/MFM/tree/main/Quantlets/Ch_01)
- ▣ Essential reading: Cont (2001); Franke, Härdle & Hafner (2019); Campbell, Lo & MacKinlay (1997), ch. 1–2

## References (1/4)

- Andersen, T.G., Bollerslev, T., Diebold, F.X., Ebens, H. (2001). The distribution of realized stock return volatility. *Journal of Financial Economics*, 61(1), 43–76.
- Bachelier, L. (1900). *Théorie de la spéculation*. *Annales scientifiques de l'École Normale Supérieure*, 17, 21–86.
- Bai, J., Ng, S. (2005). Tests for skewness, kurtosis, and normality for time series data. *Journal of Business & Economic Statistics*, 23(1), 49–60.
- Bollerslev, T., Hood, B., Huss, J., Pedersen, L.H. (2018). Risk everywhere: modeling and managing volatility. *Review of Financial Studies*, 31(7), 2729–2773.
- Bouchaud, J.-P., Matacz, A., Potters, M. (2001). Leverage effect in financial markets: the retarded volatility model. *Physical Review Letters*, 87(22), 228701.
- Campbell, J.Y., Lo, A.W., MacKinlay, A.C. (1997). *The Econometrics of Financial Markets*. Princeton University Press.
- Christie, A.A. (1982). The stochastic behavior of common stock variances: value, leverage and interest rate effects. *Journal of Financial Economics*, 10(4), 407–432.
- Clark, P.K. (1973). A subordinated stochastic process model with finite variance for speculative prices. *Econometrica*, 41(1), 135–155.
- Cong, L.W., Li, X., Tang, K., Yang, Y. (2023). Crypto wash trading. *Management Science*, 69(11), 6427–6454.
- Cont, R. (2001). Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1(2), 223–236.
- Dalla, V., Giraitis, L., Phillips, P.C.B. (2022). Robust tests for white noise and cross-correlation. *Econometric Theory*, 38(5), 913–941.
- Davis, R.A., Mikosch, T. (1998). The sample autocorrelations of heavy-tailed processes with applications to ARCH. *Annals of Statistics*, 26(5), 2049–2080.
- Diebold, F.X., Inoue, A. (2001). Long memory and regime switching. *Journal of Econometrics*, 105(1), 131–159.
- Ding, Z., Granger, C.W.J., Engle, R.F. (1993). A long memory property of stock market returns and a new model. *Journal of Empirical Finance*, 1(1), 83–106.

## References (2/4)

- Drees, H., Resnick, S., de Haan, L. (2000). How to make a Hill plot. *Annals of Statistics*, 28(1), 254–274.
- Engle, R.F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.
- Fama, E.F. (1965). The behavior of stock-market prices. *Journal of Business*, 38(1), 34–105.
- Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- Gabaix, X., Ibragimov, R. (2011). Rank –  $1/2$ : a simple way to improve the OLS estimation of tail exponents. *Journal of Business & Economic Statistics*, 29(1), 24–39.
- Garman, M.B., Klass, M.J. (1980). On the estimation of security price volatilities from historical data. *Journal of Business*, 53(1), 67–78.
- Geweke, J., Porter-Hudak, S. (1983). The estimation and application of long memory time series models. *Journal of Time Series Analysis*, 4(4), 221–238.
- Granger, C.W.J., Ding, Z. (1995). Some properties of absolute return: an alternative measure of risk. *Annales d'Économie et de Statistique*, 40, 67–91.
- Guyon, J., Lekeufack, J. (2023). Volatility is (mostly) path-dependent. *Quantitative Finance*, 23(9), 1221–1258.
- Härdle, W.K., Chen, C.Y.-H., Overbeck, L. (Eds.) (2017). *Applied Quantitative Finance*, 3rd ed. Springer.
- Härdle, W.K., Harvey, C.R., Reule, R.C.G. (2020). Understanding cryptocurrencies. *Journal of Financial Econometrics*, 18(2), 181–208.
- Hill, B.M. (1975). A simple general approach to inference about the tail of a distribution. *Annals of Statistics*, 3(5), 1163–1174.
- Hoga, Y. (2017). Change point tests for the tail index of  $\beta$ -mixing random variables. *Econometric Theory*, 33(4), 915–954.
- Hsing, T. (1991). On tail index estimation using dependent data. *Annals of Statistics*, 19(3), 1547–1569.

## References (3/4)

- Inclán, C., Tiao, G.C. (1994). Use of cumulative sums of squares for retrospective detection of changes of variance. *Journal of the American Statistical Association*, 89(427), 913–923.
- Jarque, C.M., Bera, A.K. (1980). Efficient tests for normality, homoscedasticity and serial independence of regression residuals. *Economics Letters*, 6(3), 255–259.
- Karpoff, J.M. (1987). The relation between price changes and trading volume: a survey. *Journal of Financial and Quantitative Analysis*, 22(1), 109–126.
- Kim, T.-H., White, H. (2004). On more robust estimation of skewness and kurtosis. *Finance Research Letters*, 1(1), 56–73.
- Ledoit, O., Wolf, M. (2008). Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance*, 15(5), 850–859.
- Ljung, G.M., Box, G.E.P. (1978). On a measure of lack of fit in time series models. *Biometrika*, 65(2), 297–303.
- Lo, A.W. (2002). The statistics of Sharpe ratios. *Financial Analysts Journal*, 58(4), 36–52.
- Lobato, I., Nankervis, J.C., Savin, N.E. (2001). Testing for autocorrelation using a modified Box–Pierce Q test. *International Economic Review*, 42(1), 187–205.
- Magdon-Ismail, M., Atiya, A.F., Pratap, A., Abu-Mostafa, Y.S. (2004). On the maximum drawdown of a Brownian motion. *Journal of Applied Probability*, 41(1), 147–161.
- Mandelbrot, B. (1963). The variation of certain speculative prices. *Journal of Business*, 36(4), 394–419.
- Mikosch, T., Stărică, C. (2004). Nonstationarities in financial time series, the long-range dependence, and the IGARCH effects. *Review of Economics and Statistics*, 86(1), 378–390.
- Opdyke, J.D. (2007). Comparing Sharpe ratios: so where are the p-values?. *Journal of Asset Management*, 8(5), 308–336.
- Parkinson, M. (1980). The extreme value method for estimating the variance of the rate of return. *Journal of Business*, 53(1), 61–65.

## References (4/4)

- ▣ Pele, D.T. (2014). [A SAS approach for estimating the parameters of an alpha-stable distribution](#). *Procedia Economics and Finance*, 10, 68–77.
- ▣ Pele, D.T., Lazar, E., Dufour, A. (2017). [Information entropy and measures of market risk](#). *Entropy*, 19(5), 226.
- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). [Are cryptos becoming alternative assets?](#). *The European Journal of Finance*, 29(10), 1064–1105.
- ▣ Praetz, P.D. (1972). [The distribution of share price changes](#). *Journal of Business*, 45(1), 49–55.
- ▣ Ratliff-Crain, E., Van Oort, C.M., Koehler, M., Tivnan, B.F. (2025). [Revisiting Cont's stylized facts for modern stock markets](#). *Quantitative Finance*, 25(9), 1343–1373.
- ▣ Robinson, P.M. (1995). [Gaussian semiparametric estimation of long range dependence](#). *Annals of Statistics*, 23(5), 1630–1661.
- ▣ Rogers, L.C.G., Satchell, S.E. (1991). [Estimating variance from high, low and closing prices](#). *Annals of Applied Probability*, 1(4), 504–512.
- ▣ Romano, J.P., Thombs, L.A. (1996). [Inference for autocorrelations under weak assumptions](#). *Journal of the American Statistical Association*, 91(434), 590–600.
- ▣ Sharpe, W.F. (1966). [Mutual fund performance](#). *Journal of Business*, 39(1), 119–138.
- ▣ Sharpe, W.F. (1994). [The Sharpe ratio](#). *Journal of Portfolio Management*, 21(1), 49–58.
- ▣ Sortino, F.A., van der Meer, R. (1991). [Downside risk](#). *Journal of Portfolio Management*, 17(4), 27–31.
- ▣ Student (1908). [The probable error of a mean](#). *Biometrika*, 6(1), 1–25.
- ▣ Taylor, S.J. (1986). *Modelling Financial Time Series*. Wiley; 2nd ed. (2008), World Scientific, doi:10.1142/6578.
- ▣ Yang, D., Zhang, Q. (2000). [Drift-independent volatility estimation based on high, low, open, and close prices](#). *Journal of Business*, 73(3), 477–492.